



ANGLO-CHINESE JUNIOR COLLEGE
Preliminary Examination 2015

BIOLOGY

9648/03

Applications Paper and Planning Question

PAPER 3

**T3W10
2 hours**

Additional Materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your name, subject class, form class and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE	
1	
2	
3	
4	
5	
TOTAL	72

This document consists of **14** printed pages.

- Before the emergence of more advanced technologies, cDNA libraries have been used to study the genetic changes involved in cancer development (Fig. 1.1).

Normal cells and tumour cells are obtained from the same patient. Reverse transcription is carried out on mRNA isolated from the tumour cell with the use of primers consisting of repeats of thymine (TTTTT), forming single-stranded cDNA. These cDNA are then labelled with fluorescent dyes. Normal cell mRNA and tumour cell cDNA are allowed to hybridise; the resulting double-stranded hybrid molecules and remaining single-stranded mRNA are discarded. Non-hybridised cDNA (also known as subtracted cDNA) are used to form a subtracted cDNA library, or are used as probes to isolate genes from an existing DNA library.

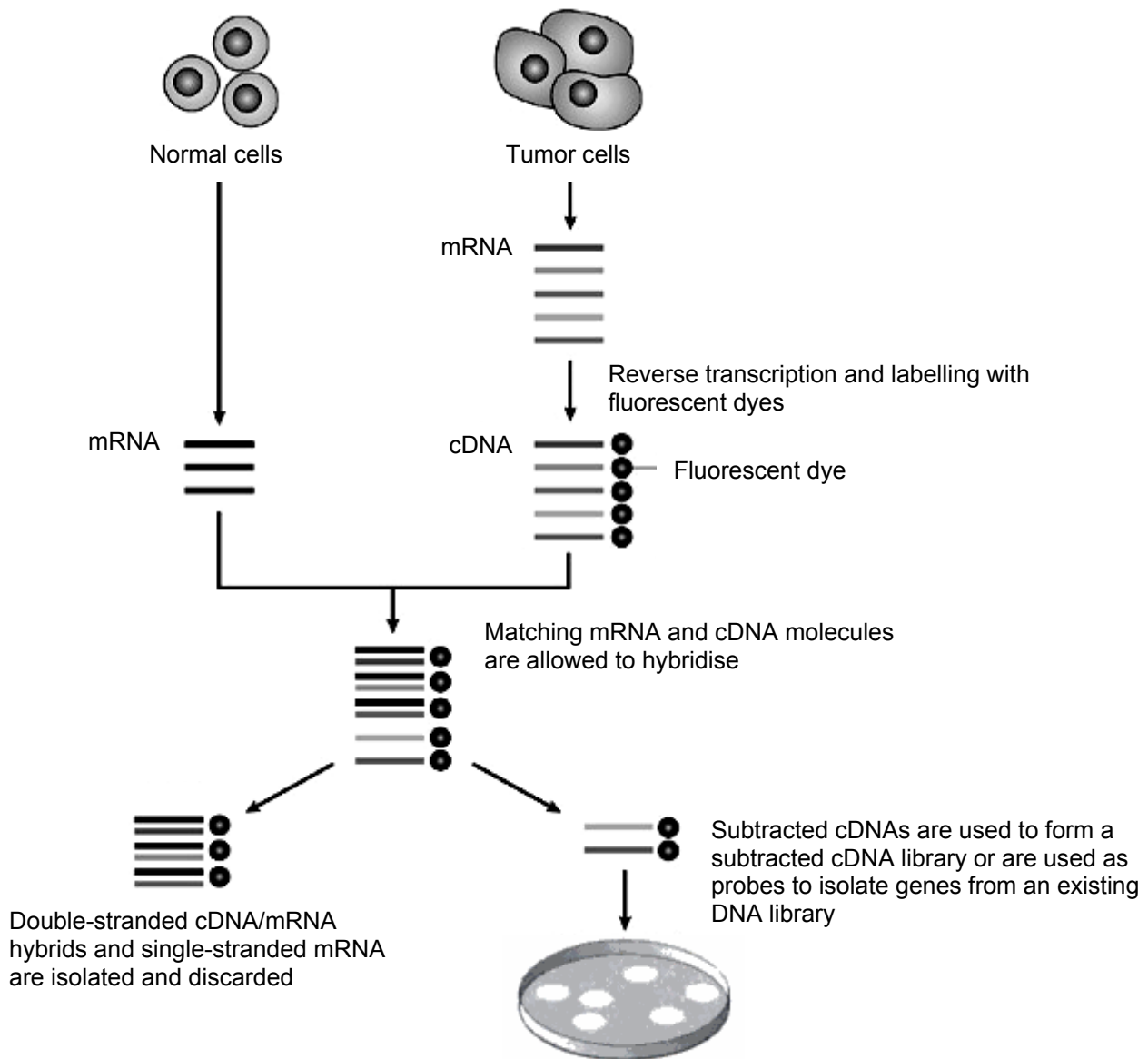


Fig. 1.1

- (a) (i) A scientist conducting this procedure was concerned that the tumour cell sample may be contaminated with bacteria. Explain why the reverse transcription of bacterial mRNA will not occur.

1. Bacterial mRNA do not possess a 3' poly(A) tail (prokaryote does not carry out post-transcriptional modification);

2. Hence oligonucleotide primer will not anneal to these mRNA, reverse transcription will not initiate;

[2]

- (ii) Describe the steps needed to form a cDNA library using subtracted cDNA and bacterial plasmids.

1. Synthesise a second strand of cDNA using DNA polymerase;

2. Linkers containing appropriate (restriction sites) are ligated to the ds cDNA (using DNA ligase);

3. cDNA and plasmids are cut/digest using the same restriction enzyme to produce sticky ends;

4. cDNA anneals with the plasmids by complementary base pairing (via H bond), and are joined using DNA ligase (which catalyses the formation of phosphodiester bonds);

5. Transformation of bacteria using plasmids by electroporation / heat shock;

6. Transformed bacteria plated onto agar plate and each transformed bacterium, (containing one cDNA fragment, forms a bacterial colony);

[5]

- (iii) Explain how the subtracted cDNA library can be used to study cancer cells.

1. Subtracted cDNA derived from tumour cells do not have complementary / anneal / binds / hybridize mRNA from normal cells;

2. hence it represents the genes that are expressed in tumour cells but not in normal cells;

3. Allows the study of differential gene expression;

4. Allows isolation / identification / studying of overexpressed genes / mutated genes / oncogenes;

5. using as radioactive gene probe (to identify cancer cell that expresses the gene);

[3]

- (iv) Suggest why the procedure described in Fig. 1.1 may not be able to detect the presence of mutated tumour suppressor genes.

1. (Mutated tumour suppressor genes) may not be expressed/ expressed in lower quantity in the tumour cell;

2. Hence no mRNA present in tumour cells to be reverse transcribed to cDNA to form subtracted cDNA/ Fewer mRNAs present in tumour cells, hence lesser cDNA formed from reverse transcription;

[2]

A DNA library can also be formed using other vectors and host cells, as shown in Table 1.1. The structure of one such vector, the yeast artificial chromosome (YAC), is shown in Fig. 1.2.

Table 1.1

Vector	Host cell	Transgene capacity
Plasmid	<i>E. coli</i>	10 kilobases
Lambda bacteriophage	<i>E. coli</i>	20 kilobases
Bacterial artificial chromosome (BAC)	<i>E. coli</i>	300 kilobases
Yeast artificial chromosome (YAC)	Yeast cells	2.5 megabases
Human artificial chromosome (HAC)	Human cells	> 2.5 megabases

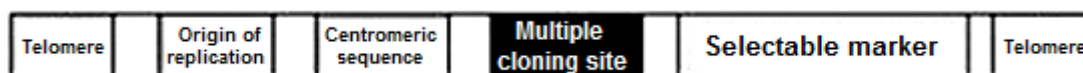


Fig. 1.2

- (b) (i) Describe **two** advantages of using YAC with yeast as host cells over using plasmids with *E. coli* as host cells, for the large-scale production of eukaryotic proteins.

1. Allows cloning of larger eukaryotic genes up to 2.5 mb;

2. YAC can be expressed in yeast cells which allow for post-translational modification of eukaryotic protein;

3. Promoter / regulatory sequence in YAC can be recognized by eukaryotic RNA polymerase / Eukaryotic transcription factors ;

4. No need for RT / steps to remove introns

[2]

- (ii) State the role of the multiple cloning site (MCS) found in YAC.

1. Contains restriction sites for several restriction enzymes, that can be digested for the insertion of the foreign gene;

[1]

[Total: 15m]

2. The first genetic maps, constructed in the early decades of the 20th century for organisms such as the fruit fly, used genes as markers. However with advancement in the understanding of genetics and molecular biology, RFLPs have been favoured as genetic markers in constructing the genetic map.

(a) Explain the advantages of RFLPs as genetic markers over genes in the construction of a genetic map.

1. Genes only occupy a small percentage of the total genome while RFLP are widespread in the genome;
2. Allows mapping in the non-coding region
3. RFLP can be found closely linked to genes (once many RFLP are found)
4. Hence genetic map constructed using genes would not be very detailed / genetic map constructed using RFLP would be more detailed
5. Genes are more conserved than RFLP / RFLP exhibit higher polymorphism than genes;
6. hence not able to differentiate between different individual / RFLP produce unique restriction fragment between different individuals;
7. Differentiation of phenotypes based on the alleles of genes would be limiting (visual or biochemical test) but clear differentiation of RFLP alleles can be done using gel E and southern blotting;

[4]

(b) In the creation of a genetic map, two RFLP loci on chromosome one are identified in fruit fly. A female heterozygous for both RFLP loci was mated with a male. The RFLP banding patterns of the mating parents as well as the F₁ offspring are shown in Fig. 2.1

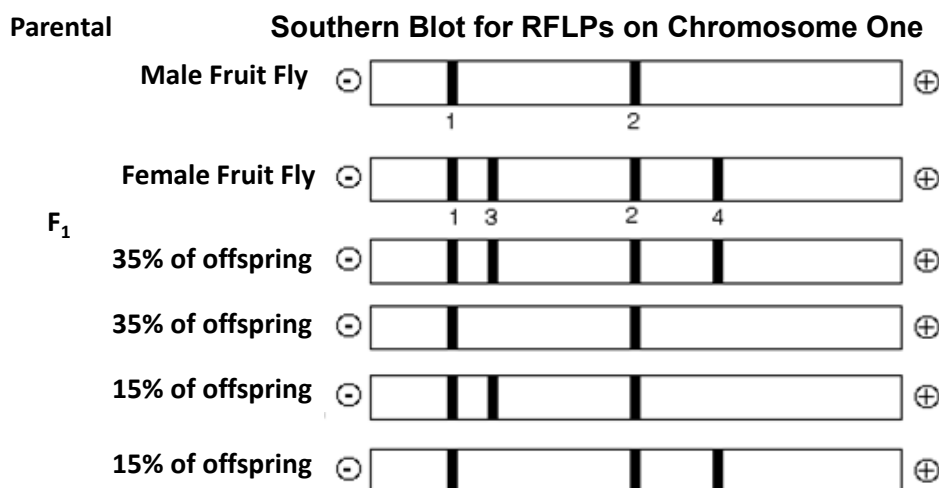


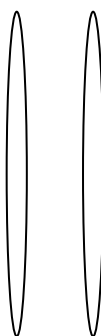
Fig. 2.1

(i) Explain why a female heterozygous for both RFLP loci was used in this experiment.

1. (Heterozygous individual at both loci would yield) <u>recombinant</u> gametes with /different band patterns via (crossing over);	(Homozygous individual at both loci would yield) <u>recombinant</u> gametes with same band patterns even via (crossing);
2. Hence able to calculate recombinant frequency to find	Hence unable to calculate recombinant frequency to find

genetic distance between the 2 locus	genetic distance between the 2 locus
[2]	

- (ii) Draw the positions of the two RFLP loci of the female fruit fly from the parental generation on the pair of homologous chromosomes below, and indicate the genetic distance between the two loci. (**distance of 30 c.m / m.u. 1 and 2 linked + 3 and 4 linked**)
[1]



- (iii) The fragments of DNA were separated using gel electrophoresis in this experiment.

Suggest and explain how increasing the concentration of the agarose during the preparation of the gel would affect the way in which DNA fragments are separated.

1. Agarose act as a molecular sieve / for the separation of the DNA fragment according to size;
 2. Increasing the concentration of agarose will decrease the size of the pores in agarose gel; (OWTTE)
 3. Hence smaller DNA fragment would be separated with higher resolution / more distinctly separated / OWTTE;
 4. Larger DNA fragment would be separated with lower resolution / less distinctly separated / OWTTE; @ 1m
- [3]

- (c) Simple sequence length polymorphism (SSLP) is another type of genetic marker that is commonly used in genetic mapping. SSLPs are repeated sequences in intergenic regions of DNA. The repeated sequences in SSLP are flanked by conserved regions. An example of SSLP is shown in Fig. 2.2.

```
...gatttctaagCAAtcttttaaag...
...gatttctaagCACAtcttttaaag...
...gatttctaagCACACAtcttttaaag...
...gatttctaagCACACACACAtcttttaaag...
...gatttctaagCACACACACACAtcttttaaag...
```

Fig. 2.2

Explain one way in which nucleic acid hybridization can be employed to distinguish between SSLP of different lengths without the use of restriction enzymes.

1. **Forward and reverse primers anneal to flanking SSLP / bind to conserved regions;**
2. **Via complementary base pairing by hydrogen bonds (can award if found in pt 5);**
3. **Amplification of repeated regions to yield fragments of different lengths via Polymerase Chain Reaction;**
4. **(DNA fragments of different lengths) separated by gel electrophoresis;**
5. **Radioactive probe (complementary to flanking conserved regions) to bind to fragment for visualization in autoradiography. / visualize with ethidium bromide**

Only award pt 4 and 5 if amplification / PCR is given;

[4]

[Total: 14m]

3. The most common treatment for severe combined immunodeficiency (SCID) is bone marrow transplantation. In the 1990s, gene therapy has been attempted as an alternative to bone marrow transplantation.

(a) (i) Explain how mutations to the interleukin-2 receptor gamma (IL2RG) gene result in symptoms exhibited in X-linked SCID patients.

1. Mutations can result change in 3D conformation of common gamma chain subunit of the protein receptor on the surface of immature blood stem cells/

2. No functional / no (common) gamma chain subunit synthesized;

3. Important chemical signals cannot be relayed to the nucleus to direct the development / maturation of T cells / B cells and natural killer cells of the immune system from blood stem cells;

4. Patients have no immunity and hence susceptible to infections;

[3]

(ii) Explain one factor that prevents the viral gene delivery method from becoming an effective long - term treatment in SCID patients.

1. It is difficult/unable to ensure integration of the normal allele of the gene into the host cell's genome;

2. Gene may be degraded by enzymes;

3. The target cells in which the normal allele is inserted into via gene therapy might be non-dividing / somatic cells which have limited number of divisions;

4. Hence do not pass on the normal allele to daughter cells/ genes not integrated are lost during cell division;

5. Risk of the viral delivery system being targeted by the immune system;

6. Hence subsequent treatment becomes ineffective; @ 1m, max 2

[2]

Note Pair pt 1/2/3 with Pt 4 or pair 1 & 2

R! normal gene, penalized once

Recently, it has been proposed that gene therapy can also be used to target cancer cells. Fig. 3.1 shows the process of how a normal white blood cell removed from a lung cancer patient is treated ex vivo with a gene coding for P receptors that specifically recognise proteins on cancer cells.

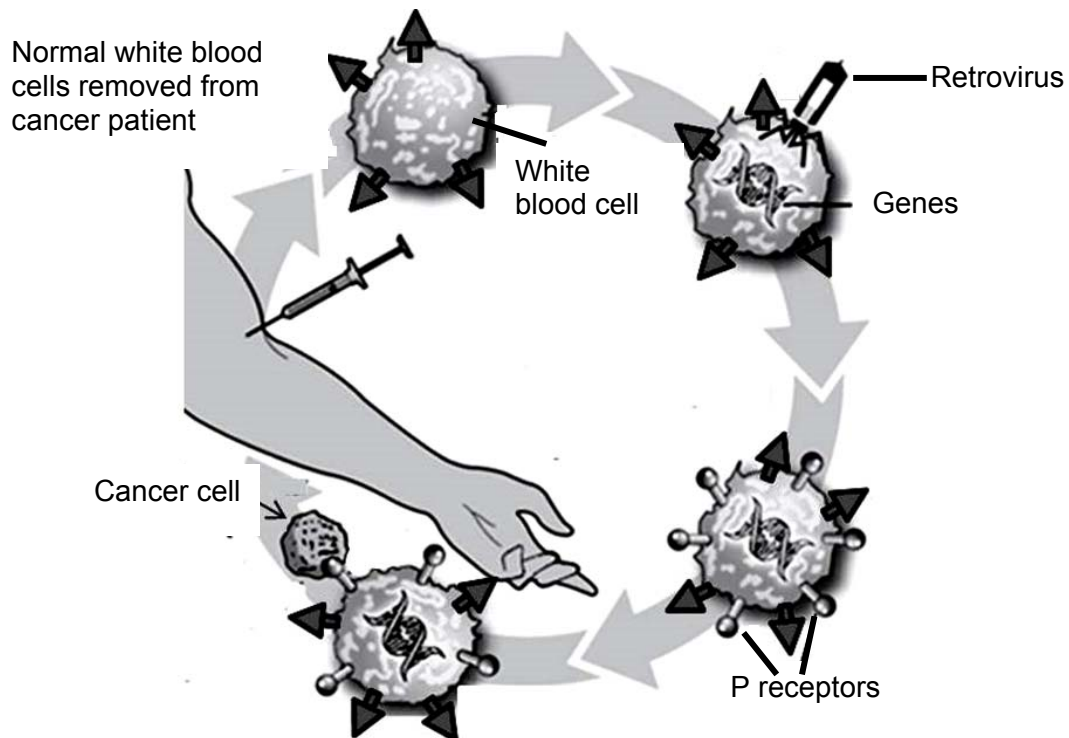


Fig. 3.1

(b) (i) With reference to Fig. 3.1, outline the method of gene therapy used to target cancer cells for degradation.

1. Gene coding for P receptors is (cloned) and inserted into inactivated/ non-pathogenic retrovirus;
2. Cell Culture: Normal white blood cells removed from cancer patient are cultured in lab;
3. Infection: Retrovirus binds to (specific receptors) on white blood cells and gene (for P receptor) transferred / transduced into the cell nucleus.
4. Gene integrated into host cell genome;
5. Cells with P receptors expressed are selected and transfused back into cancer patient;
6. P receptors of white blood cells recognise and bind to cancer cells;

[4]

(ii) Discuss the ethical considerations for such novel methods of gene therapy in targeting cancer cells.

1. Concerns about the safety/ Unknown risks of such novel methods of gene therapy to humans;
2. Due to the inadequate understanding of biological interactions of retrovirus and host/ white blood cell may recognise and target healthy cells (autoimmune disease) OWTTE ;
3. Concerns over whether patients and their family members fully understand the risks associated / whether they make informed decisions before participating in this treatment;
4. Raises questions over who should receive this treatment given the unknown

risk

- The terminally ill cancer patient or patients in early stages of cancer;
- Those who cannot afford treatment or any cancer patient who can afford the treatment;

5. White blood cell contain gene that are not naturally found in human / OWTTE

R! "tampering with nature"/ animal rights (too vague and not specific);
adult body but not those of the extra-embryonic membranes; @ @1m, max 2 [2]

[Total: 11m]

4. Planning question

The enzyme β -galactosidase hydrolyses the sugar lactose into glucose and galactose that can be absorbed into the blood stream. However, it is not easy to measure β -galactosidase activity using the appearance of glucose and galactose.

Iodine binds to a site on β -galactosidase other than the active site and changes the conformation of the active site.

In order to measure enzyme activity of β -galactosidase, an artificial substrate called o-nitrophenyl-beta-galactopyraniside (ONPG) is used. The enzyme degrades ONPG to produce beta-galactose and o-nitrophenol, a yellow coloured compound. This reaction occurs optimally at pH 5.0. The reaction is stopped by adding sodium carbonate solution to increase the pH to 11.

Using this information and your own knowledge, design an experiment to investigate the effect of increasing the concentration of iodine on the rate of hydrolysis of ONPG by β -galactosidase.

You must use:

- 1.0% β -galactosidase solution,
- 0.3% iodine solution,
- distilled water,
- 5mM ONPG
- pH 5.0 buffer,
- 0.5 M sodium carbonate solution.

You should select from the following apparatus and use appropriate additional apparatus:

- normal laboratory glassware e.g. test-tubes, beakers, measuring cylinders, glass rods etc.,
- timer e.g. stopwatch or stop clock,
- thermometer,
- colourimeter,
- thermostatically controlled water bath.

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagrams, if necessary,
- identify the independent and dependent variables,

- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- show how you will record your results and the proposed layout of results tables and graphs,
- use the correct technical and scientific terms,
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 12m]

Mark Scheme**Theoretical consideration/Rationale (2 marks)**

- Theoretical consideration or rationale of the plan to justify the practical procedure, including**
 - Ref to iodine being a non-competitive inhibitor of β -galactosidase R! copy from qs.**
 - preventing β -galactosidase-ONPG complex formation or ES complex;
 - lowering the rate of β -galactosidase-ONPG complex formation
 - ONPG cannot bind to the active site of β -galactosidase
 - ONPG is no longer complementary to β -galactosidase
 - decrease frequency of effective collisions between β -galactosidase and ONPG
 - As the concentration of iodine increases, the rate of ONPG hydrolysis decreases and hence the rate of ONP production decreases**
 - The rate of ONPG hydrolysis is measured by the rate of ONP production indicated by the change of intensity of the yellow colour per unit time.**
- Use of correct terminology**
 - enzyme-substrate complex**
 - non-competitive inhibitor**
 - effective collision**

Procedure (9 marks)

- Independent variable: concentration of Iodine**
- Specify at least 5 different concentrations of Iodine (excluding control), regular intervals between each concentration**
- Dilution table (volume of Iodine, volume of water and concentration of Iodine must be indicated, total volume is the same of all concentrations of Iodine)**

Fixed/controlled variables (points 6-11 max 2) A! listing of CV

- Reaction temperature and how temperature is maintained**
- Volume of pH to be added**
- Volume of reaction mixture**
- Volume of ONPG solution**
- Volume of β -galactosidase**
- Volume of iodine solution**
- Time of reaction**

- Description of control experiment**

- (a) Iodine is replaced with an equal volume of distilled water (as long as present in dilution table)
- (b) Subject to same experimental conditions / repeating steps _ to _
- (c) Ref to rationale of control: to show that any decrease in the rate of ONPG hydrolysis is due to the presence of the iodine.
14. Equilibration of enzyme & (buffered) ONPG separately at 37°C using a thermostatically/manually controlled waterbath for a fixed time.
15. Measurement using colourimeter.
16. Ref to details of the usage of colourimeter (zeroing using cuvette with buffer)
17. Stopping the reaction using 0.5M sodium carbonate solution (until pH 11 is reached by using a pH probe).
- 18.
- (a) replicates and
- (b) at least 2 repeats
19. calculation of rate of ONPG hydrolysis in $\text{a.u. min}^{-1} / \text{min}^{-1}$ using the following equation: (mark for formula) Allow for ECF
- Average absorbance
- Reaction time
20. Table-correct headings (Iodine/%, Absorbance/au, Average rate of ONPG hydrolysis or ONP production / $\text{a.u. min}^{-1} / \text{min}^{-1}$) + at least 2 column. Allow for ECF.
21. Graph-predicted straight line / curve (negative gradient) graph with correct axes and labels with units (x-axis: Iodine concentration/% y-axis: Rate of ONPG hydrolysis/ a.u. min^{-1})
22. Labelled diagram showing the following apparatus: (A! corresponding pts labelled in diagram even if not written in steps)
- (a) Thermostatically controlled water bath
- (b) Reaction mixture in test tube

Points 3-22 max 9 marks

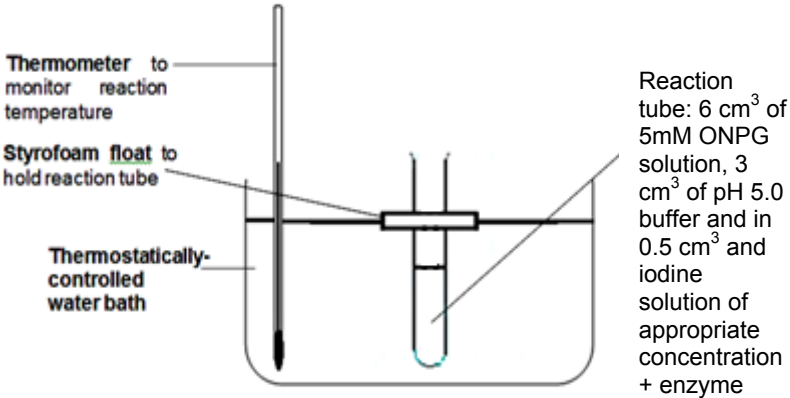
Safety precautions (1 mark)

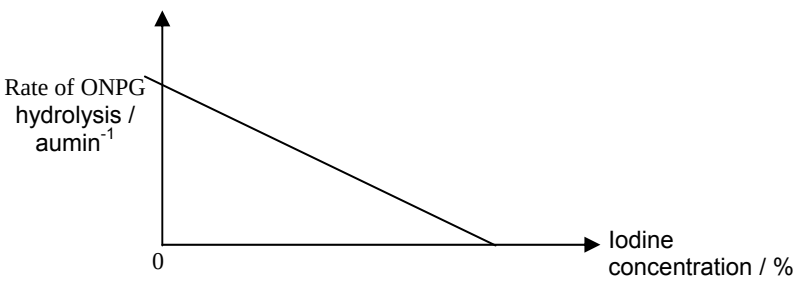
23. Two safety precautions identified
- (a) Iodine is an irritant, wear gloves
- (b) Goggles to protect hands / eyes from irritation
- (b) Do not handle the power socket of the thermostatically controlled water bath with wet hands to prevent electrocution
- (c) Handle the thermometer with caution / Do not stir the water-bath with the thermometer, to prevent breakage leading to mercury leakage and poisoning
- (d) AVP

(A) Theoretical Background and Experimental Design	
<i>Explanation of theory</i>	(a) Ref to iodine being a non-competitive inhibitor of β-galactosidase R! copy from qs. • preventing β-galactosidase-ONPG complex formation or ES complex; • lowering the rate of β-galactosidase-ONPG complex formation • ONPG cannot bind to the active site of β-galactosidase • ONPG is no longer complementary to β-galactosidase • decrease frequency of effective collisions between β-galactosidase and ONPG
<i>Hypothesis</i>	(b) As the concentration of iodine increases, the rate of ONPG

Experimental design	hydrolysis decreases and hence the rate of ONP production decreases (c) The rate of ONPG hydrolysis is measured by the rate of ONP production indicated by the change of intensity of the yellow colour per unit time.
(B) Procedure	
Variables	Independent variable: Iodine concentration Dependent variable: Rate of ONPG hydrolysis (as measured by the rate of ONP production indicated by the change of intensity of the yellow colour per unit time) Fixed / controlled variables: Reaction (ONPG hydrolysis) temperature Volume of reaction mixture Volume of pH buffer Volume of lactose solution Volume of β-galactosidase Volume of iodine solution Reaction time

Variation of IV 1. ≥ 5 equal concentration intervals, excluding control (step 1) 2. Correctly calculated volumes for both iodine and distilled water 3. Correctly calculated total volume	Variation of iodine concentration 1. Dilute the 0.3% iodine solution using appropriate volumes of distilled water as shown in the table below to obtain 2.0 cm³ iodine solutions of 6 different concentrations . <table><tr><th rowspan="2">Iodine concentration / %</th><th colspan="3">Volume / cm³</th></tr><tr><th>0.3% iodine stock</th><th>Distilled water</th><th>Total</th></tr><tr><td>0.06</td><td>0.4</td><td>1.6</td><td>2.0</td></tr><tr><td>0.12</td><td>0.8</td><td>1.2</td><td>2.0</td></tr><tr><td>0.18</td><td>1.2</td><td>0.8</td><td>2.0</td></tr><tr><td>0.24</td><td>1.6</td><td>0.4</td><td>2.0</td></tr><tr><td>0.30</td><td>2.0</td><td>0.0</td><td>2.0</td></tr></table> A: Any total volume, A: tables that include components in reaction mixture	Iodine concentration / %	Volume / cm ³			0.3% iodine stock	Distilled water	Total	0.06	0.4	1.6	2.0	0.12	0.8	1.2	2.0	0.18	1.2	0.8	2.0	0.24	1.6	0.4	2.0	0.30	2.0	0.0	2.0
Iodine concentration / %	Volume / cm ³																											
	0.3% iodine stock	Distilled water	Total																									
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0.18	1.2	0.8	2.0																									
0.24	1.6	0.4	2.0																									
0.30	2.0	0.0	2.0																									
Preparation of iodine and buffered ONPG 1. Equilibration of iodine at 37°C using thermostatically/ manually controlled water-bath 2. Controlling equilibration time (1-15 mins) using stopwatch 3. Use of appropriate volumes of ONPG, pH 5.0 buffer and iodine 4. Equilibration of buffered ONPG at 37°C using thermostatically/	Preparation of iodine solution 2. Equilibrate each iodine solution to 37 °C in the thermostatically-controlled water bath for 5 minutes . Use a stopwatch to monitor the equilibration time. Preparation of buffered ONPG solution 3. Label 5 test-tubes based on iodine concentration. 4. Add 6 cm³ of 5mM ONPG solution and 3 cm³ of pH 5.0 buffer using a 10 cm³ syringe into each test-tube. 5. Equilibrate the buffered ONPG solution in each labelled test-tube to 37 °C in the thermostatically-controlled water bath for 5 minutes . Use a stopwatch to monitor the equilibration time.																											

manually controlled water-bath	
Reaction and determination of DV 1. Concentration of enzyme solution 2. Stopping of reaction 3. Use of colourimeter	<u>Reaction</u> 6. Prepare the reaction mixture by adding in 0.5 cm³ of 0.06% iodine solution to the buffered ONPG solution using a 1 cm³ syringe and then add 1 cm³ of 1.0% β-galactosidase solution using a 1 cm³ syringe. 7. Shake the test tube to mix the contents thoroughly. 8. <i>Return the test-tube to the water-bath immediately.</i> 9. <i>Monitor the reaction for 5 minutes using the stopwatch.</i> 10. After 5 min, stop the reaction by adding 3cm³ of 0.5 M sodium carbonate solution. 11. Take out 1ml of the reaction mixture using a 1 cm³ syringe and place in into the colorimeter which has been zeroed. 12. Use, the colourimeter to determine the intensity of the yellow colour.
Drawing of setup Accept labeled drawing showing thermostatically controlled water bath, reaction mixture in test tube (thermometer)	<u>Drawing of set-up</u>  Thermometer to monitor reaction temperature Styrofoam float to hold reaction tube Thermostatically-controlled water bath Reaction tube: 6 cm³ of 5mM ONPG solution, 3 cm³ of pH 5.0 buffer and in 0.5 cm³ and iodine solution of appropriate concentration + enzyme
Reliability 1. 3 replicates 2. 3 repeats	<u>Replicates and Repeats</u> 13. Perform two more replicates by repeating steps 6 to 12 for iodine concentration: 0.06%. 14. Repeat steps 6 to 12 for the remaining iodine concentrations. 15. Repeat the entire experiment twice.
Control 1. Define control 2. Same experimental conditions (step 3. Decrease in rate	<u>Control</u> 16. Conduct the (negative) control experiment, under the same experimental conditions except that the iodine solution is replaced with an equal volume of distilled water.(A! if shown in dilution table) This shows that any decrease in the rate of ONPG hydrolysis is due to the presence of the iodine.

of ONPG hydrolysis due to iodine																																									
Data Manipulation 1. Calculate average 2. Determine rate	<u>Data Manipulation</u> 17. Calculate the average absorbance of ONP produced for the 3 replicates. 18. Calculate the rate of ONPG hydrolysis in aumin⁻¹ / min⁻¹ using the following equation: Average absorbance Reaction time where reaction time is 5 mins																																								
Results table 1. Correct IV and DV headings with suitable units 2. Properly constructed tables with complete table outline Note: Reject table if table outlines are not / incompletely draw	<u>Table showing the effect iodine/ % on the rate of ONPG hydrolysis / aumin⁻¹ / min⁻¹</u> <table><tr><th rowspan="2">Iodine concentration / %</th><th colspan="4">Absorbance/au</th><th rowspan="2">Rate of ONPG hydrolysis /aumin⁻¹</th></tr><tr><th>1</th><th>2</th><th>3</th><th>Ave</th></tr><tr><td>0.06</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>0.12</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>0.18</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>0.24</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>0.30</td><td></td><td></td><td></td><td></td><td></td></tr></table> R: volume of precipitate / colour of precipitate	Iodine concentration / %	Absorbance/au				Rate of ONPG hydrolysis /aumin ⁻¹	1	2	3	Ave	0.06						0.12						0.18						0.24						0.30					
Iodine concentration / %	Absorbance/au				Rate of ONPG hydrolysis /aumin ⁻¹																																				
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Sketch of graph 1. Correct axes labels exclude units 2. Correct trend (decreasing trend, A: linear / non-linear)	18. Plot a graph to show the effect of iodine concentration / % on the rate of ONPG hydrolysis / aumin ⁻¹ . <u>Graph showing the effect of iodine concentration / % on the rate of ONPG hydrolysis / aumin⁻¹</u>  A: graph not touching Y axis																																								

(C) Risk Assessment

Risk Assessment Note: Any risks described must be relevant to the experiment and should not be normal laboratory risks.	1. Iodine is an irritant, wear gloves 2. Wear goggles to protect hands / eyes from irritation ;; 3. Do not handle the power socket of the thermostatically controlled water bath with wet hands to prevent electrocution ;; 4. Handle the thermometer with caution / Do not stir the water-bath with the thermometer, to prevent breakage leading to mercury leakage and poisoning ;; 5. AVP
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5. Free-response question

Write your answers to this question on the separate paper provided.

Your answers:

- should be illustrated by large, clearly labelled diagrams, where appropriate.
 - must be in continuous prose, where appropriate.
 - must be set out in sections **(a)**, **(b)** etc., as indicated in the question.
- (a)** Explain how the findings of the Human Genome Project have been used in the advancement of healthcare and describe the ethical issues related to such use. [8]
- (b)** Comment on the statement “GMO can make a relevant contribution to solving the problem of food shortage in developing countries”. [6]
- (c)** With reference to a named genetic disease, explain the features of stem cell transplant that allows it to be an effective treatment. [6]

[Total: 20m]

Essays

- (a)** Explain how the findings of the Human Genome Project have been used in the advancement of healthcare and describe the ethical issues related to such use. [8]
- **Genetic testing (max 2)**
 - 1 The HGP has allowed for discovery of genes/alleles/polymorphisms associated with human diseases/ understanding genetics basis of disease;
 - 2 This allows for improved diagnosis of disease / genetic testing;
 - 3 Earlier detection of genetic predispositions to disease (eg. breast cancer, cystic fibrosis, Alzheimer’s disease);
 - 4 It also allows scientists to design drugs to target a specific gene/protein (ie. rational drug design);
 - **Pharmacogenetics**
 - 5 The HGP allows scientists/doctors to know which genes/alleles/polymorphisms affect a person’s response to a drug / genetic differences affect the way we react to a drug;
 - 6 It is (theoretically) possible to tailor drugs/treatments to fit patient’s genome for greater efficacy / avoiding dangerous side effects;
 - **Gene therapy**
 - 7 (Since gene sequences are now readily available and we better understand which genes/alleles/polymorphisms are associated with which diseases,) it is possible to use gene therapy to treat certain diseases;
 - **Risk assessment of individuals upon exposure to toxic agents**
 - 8 (The HGP has allowed for discovery of genes/alleles/polymorphisms associated with resistance/susceptibility to radiation/carcinogens, and hence) an individual’s genome/polymorphisms can be used as a means of risk assessment to evaluate health risks of individuals who may be exposed to radiation or carcinogens;

Max 4 marks for benefits

- Privacy and confidentiality of genetic information.
- 9 The issue of who owns (and controls) genetic information – whether the individual has complete control over who has access to his genetic information, or is access controlled by the company/researcher who carries out the genome sequencing, or even controlled the government;**
- Fairness in the use of genetic information by insurers, employers, courts, schools, adoption agencies, and the military, among others.
- 10 The issue of how insurers/employers/courts/schools/adoption agencies/military may request for and use DNA testing/have access to personal genetic information to discriminate people based on their genomes;**
- Psychological impact and stigmatization due to an individual's genetic differences.
- 11 It is unclear how personal genetic information affects an individual and society's perceptions of that individual / how genomic information affect members of minority communities;**
- Reproductive issues including adequate informed consent for complex and potentially controversial procedures, use of genetic information in reproductive decision making, and reproductive rights.
- 12 There is an issue of whether healthcare personnel are properly counseling parents about the risks and limitations of genetic technology (eg. with regards to the reliability of the genetic test, or whether the detected condition can be treated, and to help patients anticipate and deal with options to deal with the disease, if present, and whether relatives should be informed of the condition so that they can decide whether to test for the condition as well) ;**
- 13 The reliability and usefulness of fetal genetic testing has not been verified in many cases;**
- 14 To-be parents may have to make difficult decisions of whether to terminate pregnancy due to presence of genetic disorder (especially one for which there is currently no cure or treatment for) ;**
- Uncertainties associated with gene tests for susceptibilities and complex conditions (e.g., heart disease) linked to multiple genes and gene-environment interactions.
- 15 The issue of whether testing should be performed when no treatment is available/treatment is extremely expensive and the patient cannot afford it, as diagnosing such a condition could lead to more anxiety and frustration;**
- 16 The issue of whether parents have the right to have their children tested for adult-onset diseases, as there is potential for conflict between a parent's choice and a child's welfare (eg. a parent refuses to consent to a test that is clearly in their child's best interest, or a parent who decides to pursue a genetic "enhancement" that involves significant risks for a child, or that may limit a child's life prospects);**
- 17 There is also the related issue of who has the right to determine whether newborns or others who are incapable of valid consent (eg. mentally incapacitated) should undergo genetic screening;**
- 18 The genetic tests may only indicate a probability and not a certainty of a particular polymorphism/allele being associated with a disease or condition. (There is difficulty in interpreting a positive result because some people who carry a disease-associated mutation never develop the disease.) Hence the genetic tests may not be reliable;**

Gene therapy Ethical issues**19. Eugenics/ OWTTE****20. Unaffordable by the poor OWTTE****Max 4 marks for ethical concerns**

Using suitable examples, comment on the statement "GMO can make a relevant contribution to solving the problem of food shortage in developing countries" [6]

1. **Increase yield/harvest** of crops by genetically modifying food crops/inserting genes that will result in increased size/growth rate of food animals, or by decreasing losses caused by insect pests. Compulsory pt

Atlantic salmon

2. Genetically engineer Atlantic **salmon** by the insertion of a **growth hormone gene** from a Pacific Chinook salmon and an active promoter from an ocean pout placed upstream of the growth hormone gene.
3. The expression of the growth hormone gene results in **faster** growth rate and yield of the salmon, thereby increasing the supply of salmon.

BT corn

4. Genetically engineer **maize/potato** by insertion of a **Bt-toxin gene** from the bacteria *Bacillus thuringiensis* that codes for the production of a crystalline Bt-toxin protein.
5. The **Bt-toxin protein** kills insect larvae feeding on the plant / allow plants to have **resistance** to a range of **insect** pests. Max 3m

i) Monopolistic behaviour of biotechnology companies* (max 2)

6. Biotechnology companies may practise terminator seed technology, causing second generation seeds to be sterile. This means that farmers must buy new seeds every year;
7. Scientists have also raised concerns about innovations in research that are not shared, raising fears that world food production may be dominated by a few large biotechnology companies.
8. Patent of technology;
9. This could lead to increasing dependence of developing nations on industrialised nations/Richer countries may benefit at the expense of the poorer countries;@ 1m
10. Individual can't afford to buy GMO as they are more expensive

ii) Threat to human safety (max 1)

10. Vectors used in genetic engineering contain genes for antibiotic resistance. When transgenic foods are eaten, the genes for antibiotic resistance may pass to the *E. coli* found in the gut, making them resistant to antibiotics. The genes may then be passed on to other potentially harmful bacteria in the environment;
11. GM products have unknown effects on human health and disease/the effects may only be known after long period of exposure/the development of unknown toxic/ allergenic components;
12. Allergic reactions may occur if people unknowingly consume products containing introduced genes;

iii) Threat to safety of environment (max 1)

13. The GM crops may establish themselves as weeds;
14. Possible transfer of genes/pollen from GM crops to wild-type organisms/weeds, forming weeds/superweeds;
15. GMOs are usually able to withstand environmental stress better than other organisms within the ecosystem/have unintended effects on non-target organisms. When introduced into the ecosystem, these organisms may affect the food chains in undesirable ways/upset ecological balance;
16. GMOs could lead to reduction in biodiversity, particularly so when farmers practise monoculture of GM crops;
17. Developing pesticide/herbicide-resistant plants may encourage the use of greater amounts of pesticides/herbicides. This could drive the evolution of pesticide-resistant pests or herbicide-resistant weeds;

(c) With reference to a named genetic disease, explain the features of stem cell transplant that allows it to be an effective treatment. [6]

1. Ref to SCID / CF Spell!
2. Mutations of ADA gene / interleukin 2 receptor, gamma subunit
3. Prevents differentiation/ maturation/ development of T / B cell;
4. Impede development of immune system; pt 2 – 4 max 2
5. Hematopoietic stem cell transplant (through the bone marrow transplant)
6. Is multipotent in nature;
7. Able to undergo continual mitosis to maintain constant pool of stem cell; (continual = constant pool)
8. Able to differentiate to all blood cells including T & B cells under appropriate chemical signals;
9. As it contain normal allele for ADA gene / interleukin 2 receptor, gamma subunit gene.
10. Functional protein can be expressed in sufficient quantities (for normal maturation of T & B cells)